Unadjusted, lag-adjusted, and exposure-weighted carryover specifications under differential-correlation biomarker interactions: a factorial simulation study

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Abstract

Background. Aggregated N-of-1 trials pool repeated within-patient on/off contrasts to validate predictive biomarkers. Under the multivariate-normal (MVN) differential-correlation architecture (the covariance-moderation architecture), in which the biomarker-treatment interaction is encoded in a treatment-state-dependent correlation, a companion study found that carryover, that is, residual drug effect that decays after discontinuation, costs up to roughly 31% of power in relative terms. The analyst must still decide how to represent that residual exposure in the linear predictor. Unfortunately, it is not established which representation is most robust to the ordinarily unknown pharmacokinetic decay shape.

Methods. We compare three analysis-model specifications for the drug-exposure regressor under the covariance-moderation architecture: the binary on-drug indicator, which ignores carryover altogether (Unadjusted); that indicator augmented by a lagged just-off-drug term that estimates carryover non-parametrically (Lag-adjusted, the classical crossover device); and an exposure-weighted continuous indicator that commits to a parametric decay and half-life (Exposure-weighted, the current default). We embed them in an ADEMP-structured factorial simulation[1] crossing two data-generating process (DGP) decay forms (exponential, Weibull) with the three specifications, under three trial designs, two sample sizes, and three interaction strengths (216 cells, 500 replicates each), together with four single-axis sensitivity blocks and a cluster-robust recalibration.

All three specifications are fitted to the same simulated datasets, so every contrast between them is paired. We report each performance measure with its Monte Carlo standard error (MCSE).

Results. Two findings stand out. First, the lagged-treatment term buys nothing: Lag-adjusted and Unadjusted are statistically indistinguishable across all 216 cells, differing by a mean of 0.003 in power against an MCSE of 0.018, and agreeing to three decimal places on bias, mean squared error, and coverage. Because the two share an estimand and differ by exactly one nuisance term, this is a clean null. Second, Exposure-weighted attains the highest power only in the two designs with a blinded-discontinuation phase (Hybrid and OL+BDC), where its advantage is about 10 percentage points (0.860 vs 0.766 for Unadjusted at the reference cell); under the classical crossover (CO) design it is markedly inferior (0.488 vs 0.830). Where it leads, the lead is robust to decay-shape and half-life mis-specification and to autocorrelation strength. The model-based interaction test is mildly conservative for all three (mean null rejection 0.036 to 0.039), an artifact of the working covariance that a CR2 cluster-robust standard error removes without disturbing the ranking.

Conclusions. Three points warrant emphasis. First, under the covariance-moderation architecture we recommend the exposure-weighted predictor, reported with a cluster-robust standard error, for the Hybrid and OL+BDC designs; under a crossover design it should not be used, and the unadjusted binary indicator is materially better. Second, the classical lagged-treatment remedy cannot be recommended in any design examined here, since it never improved on ignoring carryover altogether; we trace this to its entering the model as a nuisance main effect, which leaves the attenuation of the interaction untouched. Third, across all conditions anticipated dropout governs power far more than the choice among specifications, and should dominate design effort accordingly. This manuscript restricts the grid to the covariance-moderation architecture and to the exponential and Weibull decay forms; the mean-moderation architecture and linear decay are evaluated in the archived companion drafts `archive/report.Rmd` and `archive/report-slim.Rmd`.

1 Introduction

In a single N-of-1 trial, one patient is crossed repeatedly between active treatment and a comparator over a sequence of measurement occasions, and the within-patient contrast between the on-drug and off-drug periods estimates that patient's individual treatment effect. An aggregated N-of-1 trial pools many such single-patient series into a mixed model, trading some of the resolution available within a single patient for population-level inference and for sample sizes considerably smaller than a parallel-group trial would require[2]. The estimand of interest is the biomarker-treatment interaction, that is, whether a baseline covariate B_i modifies the treatment effect. We examine three trial designs that differ in how the on-drug and off-drug occasions are arranged, a classical crossover (CO), an open-label design followed by blinded discontinuation (OL+BDC), and a Hybrid N-of-1 design; each is defined in Section 2.2.

The inferential complication specific to this setting is carryover. When a drug is discontinued, its pharmacological effect does not switch off abruptly but decays over days or weeks, so that observations recorded during the nominally off-drug period still carry residual exposure. Treating those observations as though they were cleanly off-drug mis-specifies the design matrix, and the consequences for the interaction test are worth setting out explicitly.

The exposure regressor the analyst writes down is, in that case, a mis-measured version of the exposure the patient actually received, and the damage is of two kinds. First, the contrast that the interaction term is meant to estimate is diluted. Occasions that still carry a substantial fraction of the drug effect are entered with the same value as occasions that carry none, so the coefficient on the biomarker-by-exposure product is biased toward zero, in the familiar manner of a covariate measured with error. Second, those same occasions are heterogeneous in a way the model does not represent, and the unexplained variation is absorbed into the residual variance, inflating the standard error of the very coefficient that has just been attenuated. The test statistic loses on both counts, its numerator shrinking while its denominator grows, and power falls accordingly.

We note that this is a loss of power and not a loss of validity. Under the null, $c_{bm} = 0$, the covariance-moderation architecture generates no biomarker-response association at any level of residual exposure, so a mis-coded exposure regressor has no interaction to distort, and the size of the test is unaffected by the choice of exposure coding. The mild conservatism we do observe in Section 3.5 has a separate origin in the working covariance and applies equally to both specifications. Carryover mis-specification is therefore a question of efficiency rather than a source of spurious findings, and it is on that basis that we frame what follows in terms of recovered power.

But how should residual exposure be represented in the linear predictor? Three answers are available, and the question we take up here is which of them recovers more of the power that carryover would otherwise cost. They correspond to the three positions an analyst can take toward the unknown decay: deny it, and code exposure as strictly on or off; estimate it, by adding a nuisance term whose magnitude the data determine; or assume it, by committing to a decay curve and a half-life.

A companion study[3], building on Hendrickson[2], shows that under what we call the covariance-moderation architecture, a multivariate-normal differential-correlation formulation in which the interaction lives in the treatment-state-dependent covariance $\mathrm{Cor}(B_i, BR_{it}) = c_{bm} \cdot \phi(t_{sd})$, carryover erodes power directly, and the loss is strongly design-dependent: at a half-life of one week, the relative loss is roughly 31% under OL+BDC, but only 3 to 5% under the crossover and Hybrid designs. Under an alternative architecture, in which the biomarker instead moderates the mean response directly, the corresponding loss is only 1 to 3% in every design; that architecture, together with a third combined architecture, falls outside the scope of the present manuscript, which restricts attention to the covariance-moderation architecture, where the carryover problem is largest and where Hendrickson's original results were obtained. Unfortunately, that earlier work fixed the decay shape at exponential and evaluated only a single analysis specification, leaving open the question of which representation of residual exposure performs best when the true decay shape is not known with any precision, as is ordinarily the case in practice.

We compare three ways of representing drug exposure in the linear predictor. The first is the binary on-drug indicator, which ignores carryover entirely and is what an analyst obtains by default from a standard mixed model; we call it Unadjusted, and it serves as the benchmark any remedy must beat. The second augments that indicator with a lagged just-off-drug term that estimates the magnitude of carryover without assuming any particular functional form, which we call Lag-adjusted, following the classical crossover-trial device described by Jones and Kenward[4]. The third is an exposure-weighted continuous indicator that commits to a parametric decay form and half-life, which we call Exposure-weighted and which is the current default in the pmsimstats-ng package.

Including the unadjusted benchmark is what makes the comparison interpretable rather than merely competitive. Unadjusted and Lag-adjusted estimate the same coefficient and are fitted to the same simulated datasets, differing by exactly one nuisance term, so the contrast between them isolates the contribution of that term alone. Exposure-weighted differs on a second axis as well, since it changes the regressor that carries the interaction. We cross the three specifications against exponential and Weibull data-generating decay forms and ask whether one can be recommended outright, or whether the choice depends on the trial design. The study follows the ADEMP reporting framework of Morris, White, and Crowther[1], and every performance measure we report carries its Monte Carlo standard error. Section 2 describes the simulation design in ADEMP order, Section 3 gives the results, and Section 4 takes up the practical implications for trialists.

2 Simulation design

The study simulates complete aggregated N-of-1 trials and analyzes each one under every specification. A single replicate proceeds as follows. Patients are allocated across the randomization paths of the chosen trial design (Section 2.2). For each patient we draw a pre-treatment biomarker together with the three latent response components, namely the pharmacological response, the natural-history trend and the placebo-belief component, jointly across all eight measurement occasions from a multivariate normal distribution whose means follow modified Gompertz trajectories and whose covariance carries within-component AR(1) autocorrelation, cross-component correlations, and the treatment-state-dependent biomarker-response correlation that encodes the interaction (Section 2.3).

The observed symptom score at each occasion is the patient's baseline severity less the sum of the three components, so that a beneficial effect lowers the score. Carryover enters through a decay function of time since discontinuation, which keeps part of the pharmacological effect present at nominally off-drug occasions. All three analysis specifications (Section 2.4) are then fitted to that same simulated dataset, and we record the interaction coefficient, its standard error and its p-value from each. This is repeated 500 times for every parameter combination in the grid of Section 2.5.

Fitting all specifications to a common dataset is deliberate: it makes every contrast among them paired, so that differences are estimated with substantially less Monte Carlo variance than independent draws would give. Throughout, N denotes the **total** number of patients in a trial, allocated as evenly as possible across the design's randomization paths, so a design with P paths assigns about N/P patients to each. At $N = 70$ this is 35 per path under the two-path designs and 17 or 18 per path under the four-path Hybrid design.

2.1 Aims

The aim of this study is to quantify, under the covariance-moderation architecture, the cost of mis-specifying the carryover representation in the analysis model, and to determine which of the three candidate strategies, Unadjusted, Lag-adjusted or Exposure-weighted, performs better across the grid of data-generating decay forms and trial designs. A second and related aim, addressed in the Tier 2 sensitivity blocks of Section 3.6, is to check whether the resulting ranking survives perturbation of the within-subject autocorrelation, the dropout rate, the accuracy of the assumed half-life, and the size of the biomarker-treatment interaction itself.

2.2 Trial designs

The three designs differ in how the on-drug and off-drug occasions are arranged, and therefore in how much post-discontinuation information each generates. All three schedule eight measurement occasions per patient and allocate patients across randomization paths, each path being a fixed schedule of on-drug and off-drug periods. Each design also carries an expectancy weight per occasion, one during open-label periods and one half during blinded periods, which scales the placebo-belief component of the response.

Crossover (CO). The traditional two-period within-patient crossover, blinded throughout, with occasions evenly spaced at 2.5-week intervals from 2.5 to 20 weeks. One of the two paths receives active treatment for the first four occasions and comparator thereafter; the other reverses that order and so never discontinues, contributing no post-discontinuation observations at all.

Open-label followed by blinded discontinuation (OL+BDC). Patients are titrated openly on active treatment across four occasions at 4, 8, 12 and 16 weeks, then followed weekly to week 20, with a blinded switch to placebo late in the schedule. One of the two paths discontinues after week 18 and the other after week 17, so both discontinue but only two or three occasions follow.

Hybrid. The Hendrickson design and this program's reference. Open-label titration at weeks 4 and 8 is followed by blinded discontinuation across weekly occasions from week 9 to week 12, and then by a brief two-occasion crossover at weeks 16 and 20. The four paths differ in whether discontinuation begins after the third or the fourth occasion, and in which arm the patient takes first in the closing crossover. It is the richest of the three in on-off transitions and the only one in which a patient may discontinue more than once.

One consequence of these schedules bears directly on the results. The first post-discontinuation occasion falls one week after the switch under OL+BDC and Hybrid, but 2.5 weeks after it under CO. At the half-lives examined here, roughly half the pharmacological effect therefore persists into the first off-drug occasion under the two non-crossover designs, against less than a fifth under CO. The designs differ substantially, in other words, in how much carryover contamination they can accumulate at all, and Section 4.2 returns to this.

2.3 Data-generating mechanism

Under the covariance-moderation architecture, the biomarker and the biological response are drawn jointly from a multivariate normal distribution with a treatment-state-dependent correlation,

$\mathrm{Cor}(B_i, BR_{it}) = c_{bm} \cdot \phi(t_{sd})$, where ϕ denotes the decay form and t_{sd} is the time elapsed since drug discontinuation. The mean-moderation architecture, in which the biomarker instead moderates the mean response directly, and a combined the dual-channel architecture, fall outside the scope of the present manuscript; the full comparison across architectures is reported in the archived full-scope drafts (`archive/report.Rmd`, `archive/report-slim.Rmd`). In both architectures, the decay function $\phi : \mathbb{R}_+ \to [0, 1]$ maps t_{sd} to a residual-effect multiplier. We evaluate two such forms here, each parameterized to match a common half-life $t_{1/2}$. A third form, linear decay, appears in the full-scope manuscripts but is not considered further here.

2.3.1 Exponential decay

$$\phi_{\exp}(t_{sd}) = \exp\left(-\lambda t_{sd}\right), \qquad \lambda = \frac{\ln 2}{t_{1/2}}$$

This is the decay form used throughout the companion manuscript, and it is the one currently implemented in `functions.R` under `implementations/tidyverse/`. It corresponds to classical first-order elimination kinetics and is the default assumption in pharmacokinetics; we retain it here as the reference against which the Weibull form is compared.

2.3.2 Weibull decay

$$\phi_{\text{weib}}(t_{sd}) = \exp\left(-\left(\lambda_w t_{sd}\right)^k\right), \qquad \lambda_w = \frac{(\ln 2)^{1/k}}{t_{1/2}}$$

A shape parameter k governs whether the elimination tail is heavier than exponential ($k < 1$, slower elimination) or lighter ($k > 1$, faster off-drug clearance); the exponential form is recovered exactly at $k = 1$. We report results at $k = 0.7$, representing a drug with a prolonged low-concentration tail, and at $k = 1.5$, representing a drug cleared more rapidly once concentration falls below some saturation threshold, perhaps through active transport. Together with $k = 1.0$, which is numerically indistinguishable from the exponential form, this gives four decay-shape levels in the grid.

2.4 Estimand and analysis specifications

Under the covariance-moderation architecture, the interaction estimand is the on-drug biomarker-response correlation c_{bm}, a dimensionless quantity. Bias and empirical standard error are reported on a calibrated effect-size scale, namely the expected difference in on-drug response per standard deviation of the biomarker at $t_{1/2} = 0$, as defined in `docs/19-mean-moderation-implementation-notes.tex`; power and Type I error are reported on the native test-statistic scale. The null estimand used for Type I error verification is $c_{bm} = 0$, realized by the corresponding row of the parameter grid.

The analysis model is `nlme::lme(Sx ~ bm * X + t, random = ~1 | ptID, correlation = corCAR1(form = ~ t | ptID))` for all three specifications considered here; they differ only in how the drug-exposure predictor X is constructed, and in whether a nuisance term is added.

One distinction governs how the results below must be read. Unadjusted and Lag-adjusted both test the coefficient on bm:D_{it}, so they estimate the same quantity and every performance measure is comparable between them. Exposure-weighted tests the coefficient on bm:$D_{bc,it}$, a different quantity. Power and Type I error are properties of the test rather than of the coefficient tested and so remain comparable across all three, but bias, mean squared error and coverage are not comparable across that divide, for the reason given in Section 3.1.

2.4.1 Unadjusted: binary on-drug indicator

Under Unadjusted, $X = D_{it}$, the binary on-drug indicator, which takes the value one at on-drug timepoints and zero otherwise. Carryover is not represented at all: an off-drug occasion still carrying most of the pharmacological effect is coded identically to one carrying none. This is the specification an analyst obtains without considering carryover, and it enters the comparison as the benchmark against which the other two must justify their additional structure. It consumes no half-life and is therefore invariant to the analyst's pharmacokinetic assumptions.

2.4.2 Lag-adjusted: binary indicator plus estimated carryover term

Under Lag-adjusted, $X = D_{it}$ is retained and the model is augmented with a lagged-treatment indicator $L_{it} = D_{i,t-1}(1 - D_{it})$, which marks the off-drug timepoints immediately following an on-drug timepoint. The magnitude of carryover is estimated from the data rather than assumed in advance, so this specification does not require the analyst to posit any particular decay form, and like Unadjusted it is invariant to the assumed half-life. The device follows the classical crossover-trial tradition surveyed by Jones and Kenward[4].

One structural feature bears directly on how its results should be read: the lagged term L_{it} enters only as a nuisance main effect, with no bm:L_{it} product, and the interaction actually tested remains bm:D_{it}. Lag-adjusted can therefore absorb mean-level carryover, but it cannot repair the attenuation of the biomarker interaction described in Section 1. It is best understood as Unadjusted with a carryover nuisance control added, rather than as a fundamentally different model of the interaction, and Section 4.1 takes up what follows from that.

2.4.3 Exposure-weighted: continuous decayed predictor

Under Exposure-weighted, $X = \mathrm{Dbc}_{it}$, a continuous exposure-decayed predictor constructed to match the analyst's assumed decay form and half-life; this is the current pmsimstats-ng default. At on-drug timepoints $D_{bc,it} = 1$ regardless of the assumed half-life. At off-drug timepoints the predictor decays as $D_{bc,it} = \exp(-(\ln 2/\hat{t}_{1/2})\, t_{sd,it}) = (1/2)^{t_{sd,it}/\hat{t}_{1/2}}$, losing exactly half its value for every $\hat{t}_{1/2}$ weeks that elapse after discontinuation. When the analyst's assumed decay form matches the true data-generating form, the predictor is well matched; when it does not, it is mis-specified in a way we probe directly in Tier 2 block S2. We note that the half-life assumption enters only through the predictor values handed to `lme`, not through the model formula, which is identical regardless of $\hat{t}_{1/2}$. That is, Exposure-weighted is in effect a different model for every value of the assumed half-life, even though it is written the same way each time. It is the only one of the three that consumes a half-life at all, and that asymmetry, one specification indexed continuously by $\hat{t}_{1/2}$ and two that are not members of that family, is what the Tier 2 block S2

Table 1: Principal simulation grid. Cell 3 is the matched reference: exponential DGP analyzed under the current pmsimstats-ng default. Cells in the Exposure-weighted column below the first row quantify the cost of decay-form mis-specification; the Unadjusted and Lag-adjusted columns are invariant to the assumed decay form by construction.

DGP decay	Unadjusted	Lag-adjusted	Exposure-weighted
Exponential	cell 1	cell 2	cell 3 (matched)
Weibull (k = 0.7)	cell 4	cell 5	cell 6
Weibull (k = 1.0)	cell 7	cell 8	cell 9
Weibull (k = 1.5)	cell 10	cell 11	cell 12

sensitivity contrast in Section 3.6 is designed to probe.

A limiting-case relationship connects the first and third specifications and is worth stating, because it makes the set a continuum rather than three unrelated choices. As $\hat{t}_{1/2} \to 0$ the decay rate diverges and $D_{bc,it} \to D_{it}$, so Unadjusted is exactly the boundary member of the Exposure-weighted family at an assumed half-life of zero. The S2 sweep in Section 3.6 is therefore a traverse along a path anchored at Unadjusted, and not an independent robustness check.

2.5 Factorial grid and parameter settings

The principal comparison is a factorial of DGP decay form (four levels: exponential; Weibull at $k \in \{0.7, 1.0, 1.5\}$) against analysis-model specification (three levels), under the covariance-moderation architecture only:

Parameter	Values
Biomarker moderation (c_{bm})	0.0, 0.30, 0.45
Carryover half-life ($t_{1/2}$)	0.0, 0.5, 1.0 weeks
Weibull shape (k)	0.7, 1.0, 1.5 (Weibull form only)
Sample size (N)	35, 70
Trial design	CO, N-of-1 (Hybrid), OL+BDC
DGP architecture	B (MVN differential correlation) only
Replicates per cell	500 (target); 50 for development

The 216-cell count follows the same nesting logic used in the full-scope manuscript, with the architecture dimension fixed at B and the linear decay form removed. The decay-shape combinations number four (exponential, and Weibull at $k \in \{0.7, 1.0, 1.5\}$), so the grid comes to $4 \times 3 \times 3 \times 2 \times 3 = 216$ cells, the factors being decay shape, carryover half-life, trial design, sample size, and biomarker level. The $c_{bm} = 0$ row serves as the Type I error check, targeting the nominal 5% level, and the $t_{1/2} = 0$ row, in which carryover is absent altogether, gives a best-case power reference at each combination of sample size, design, and c_{bm}, against which the power cost of nonzero carryover can be read off directly.

2.6 Inference and the cluster-robust correction

The interaction test described so far is model-based: the standard error of $\beta_{bm:X}$ is read from the fitted `lme` object, and is therefore valid only insofar as the assumed residual covariance is correct. That assumption is not innocuous here. The working covariance combines a patient random intercept with a `corCAR1` continuous-time AR(1) residual correlation, and the companion calibration study[5] shows that this structure does not match the covariance the data-generating mechanism actually induces. The mismatch inflates the estimated standard error by roughly six to ten percent, and the test rejects correspondingly less often than its nominal rate.

We quantify that inflation with the calibration ratio κ, the mean estimated standard error divided by the empirical standard deviation of the interaction estimate across replicates. A correctly calibrated test has $\kappa = 1$; $\kappa > 1$ means the standard error is too large and the test is conservative; $\kappa < 1$ means it is anti-conservative. At the null cell we observe κ between 1.05 and 1.10 across the three specifications, with empirical Type I error between 0.029 and 0.039 against a nominal 0.05.

The remedy is a cluster-robust, or sandwich, variance estimator. Such an estimator derives the variance of $\hat{\beta}_{bm:X}$ from the observed between-cluster variation in the model's estimating-equation contributions rather than from the assumed covariance, and so remains valid when the working covariance is mis-specified. The clusters are patients, each contributing eight measurement occasions; patient identifiers are made unique across randomization paths before clustering, so no cluster spans two paths.

Two features of this setting dictate which sandwich estimator is appropriate. First, the naive form is markedly downward-biased when the number of clusters is small, and with $N = 35$ or $N = 70$ patients that bias is not negligible. We therefore use the CR2 bias-reduced estimator of Bell and McCaffrey, which applies a cluster-specific adjustment chosen so that the variance estimate is exactly unbiased under a working model. Second, with few clusters the resulting statistic is not well approximated by a normal reference distribution, so we pair CR2 with Satterthwaite degrees of freedom estimated per cluster configuration; these come out well below the naive residual count and are reported per cell alongside the results. Computation uses the `clubSandwich` package, applied to each specification separately: the fitted `lme` object is retained and only its standard error, reference distribution and p-value are replaced, so the point estimate is untouched and the comparison isolates the effect of the inference procedure.

We apply the correction in two places, both anchored at the covariance-moderation architecture. A null-cell diagnostic ($c_{bm} = 0$, exponential DGP, $t_{1/2} = 1.0$ weeks, Hybrid, $N = 70$, 3,000 replicates) establishes whether CR2 restores nominal size, and is reported in Section 3.5. A five-cell block, comprising the reference cell and four stress cells (small N, high autocorrelation, half-life mis-specification, and 30% MCAR dropout, 500 replicates each), then asks whether restoring size disturbs the ranking among specifications; it is reported in the same section. Following this manuscript's block numbering, that five-cell block is S6. Blocks S1 through S4 are the Tier 2 single-axis sensitivity analyses of Section 3.6; block S5, an exploratory autocorrelation by carryover interaction, falls outside the scope adopted here and is reported only in the archived full-scope drafts.

2.7 Performance measures and computing environment

For each cell we report Power, Type I error (at $c_{bm} = 0$), Bias and Empirical SE of the interaction estimate (on the calibrated scale), Coverage of the nominal 95% Wald CI, MSE, the non-convergence rate, and the MCSE of each measure, following Morris et al.[1, Tables 5-6].

We chose $n_{sim} = 500$ so that the Monte Carlo standard error on power at $p = 0.5$ would not exceed 0.022 ($\sqrt{0.5 \cdot 0.5/500} \approx 0.022$).

No simulation is run for this manuscript; we instead filter the same production-scale grid summary used in the archived full-scope drafts (540 cells, `analysis/data/02-grid-summary.rds`, computed with `implementations/tidyverse/` as the primary implementation under R 4.5.3, 128 minutes on eight cores, commit SHA `7018b59`, dated 2026-04-15) down to the 216-cell slice defined by the covariance-moderation architecture and the two non-linear decay forms. Within each cell, all three specifications were fit to the same simulated dataset (common random numbers), so every pairwise contrast among them is estimated with reduced Monte Carlo variance relative to what independent draws would give. This matters most for the Lag-adjusted minus Unadjusted contrast, which differs by a single nuisance term. The Tier 2 sensitivity blocks and the S6 cluster-robust recalibration are filtered from the same shared outputs, `02-sensitivity-summary.rds` under `analysis/data/` and `diag-s6-cr2.rds` under the driver output directory; their designs were already anchored at the covariance-moderation architecture. The filtered figures used here are produced by `03-render-figures-extra-slim.R`, `04-sensitivity-figures-extra-slim.R`, and `07-type1-cr2-figure-extra-slim.R`. The Reproducibility section gives the full paths.

3 Results

All results reported below derive from the same 500-replicate production run used in the full-scope manuscripts, filtered to the covariance-moderation architecture and the exponential and Weibull decay forms. The Monte Carlo standard error on power does not exceed 0.022 at $p = 0.5$, and differences smaller than this should not be over-interpreted. Coverage of the nominal 95% Wald confidence interval is reported for the reference cell, and non-convergence rates are reported per cell in the tables that follow.

3.1 Headline performance table

Table 2 reports performance at the reference cell (the covariance-moderation architecture, exponential DGP, Hybrid design, $N = 70$, $c_{bm} = 0.45$), where non-convergence was zero throughout. A caveat applies to the bias, mean squared error, and coverage columns:

Table 2: Headline performance at the reference cell (exponential DGP, Hybrid design, $N = 70$, $c_{bm} = 0.45$, $n_{sim} = 500$). Power is the rejection rate of $H_0 : \beta_{bm:X} = 0$ at $\alpha = 0.05$. Bias is on the calibrated effect-size scale, $\theta_{\text{true}} = -c_{bm} \cdot \sigma_{BR} = -3.6$ (per docs/19 calibration). MCSEs are reported per Morris et al. (2019, Table 6).

t1/2	Specification	Power (MCSE)	Bias (MCSE)	Empirical SE	Coverage (MCSE)	MSE (MCSE)	Non-conv
0.0	Unadjusted	0.988 (0.005)	-0.029 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.0	Lag-adjusted	0.990 (0.004)	-0.025 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.0	Exposure-weighted	0.988 (0.005)	-0.029 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.5	Unadjusted	0.918 (0.012)	+0.586 (0.036)	0.807	0.938 (0.011)	0.995 (0.058)	0.000
0.5	Lag-adjusted	0.920 (0.012)	+0.583 (0.036)	0.812	0.938 (0.011)	1.000 (0.059)	0.000
0.5	Exposure-weighted	0.948 (0.010)	+0.072 (0.041)	0.907	0.972 (0.007)	0.828 (0.057)	0.000
1.0	Unadjusted	0.766 (0.019)	+1.093 (0.038)	0.847	0.828 (0.017)	1.911 (0.093)	0.000
1.0	Lag-adjusted	0.770 (0.019)	+1.099 (0.038)	0.853	0.818 (0.017)	1.935 (0.094)	0.000
1.0	Exposure-weighted	0.860 (0.016)	-0.081 (0.049)	1.085	0.968 (0.008)	1.184 (0.077)	0.000

these are computed against Exposure-weighted's calibrated target, $\theta_{\text{true}} = -c_{bm}\sigma_{BR} = -3.6$, while Unadjusted and Lag-adjusted estimate a different coefficient, bm:D_{it}. Their apparent bias and under-coverage in this table reflect that mismatch in estimand rather than any defect in the estimators. Those three columns are therefore comparable *within* the Unadjusted and Lag-adjusted pair, which share a target, but not across to Exposure-weighted; for the three-way comparison we rely on power and Type I error, which are properties of the test rather than of the coefficient tested.

3.2 Matched-decay baseline

At the matched cell (exponential DGP, exposure-weighted analysis, $N = 70$, $c_{bm} = 0.45$, Hybrid design), power declines from 0.988 ± 0.005 at $t_{1/2} = 0$ to 0.948 ± 0.010 at $t_{1/2} = 0.5$ to 0.860 ± 0.016 at $t_{1/2} = 1.0$ weeks (point estimate plus or minus MCSE, $n_{sim} = 500$). All 500 replicates converged at every half-life in this cell.

We had previously read the value at $t_{1/2} = 1.0$ as agreeing with the power of 0.82 reported by Hendrickson et al. for a comparable cell[2]. We withdraw that comparison. In the reference implementation, the analysis-side and data-generating carryover parameters are set independently, and the drivers that reproduce the original figures supply a half-life to the data-generating step only, leaving the analysis-side half-life at its default of zero, at which the exposure-decayed predictor collapses onto the binary on-drug indicator. The reference analysis is therefore Unadjusted, not Exposure-weighted, so the appropriate comparator is our unadjusted power at the same cell, 0.766, and the grids differ further on interaction strength and half-life, so "the same cell" was approximate in any case. We have not resolved what the published figure was computed under, and make no claim of agreement or disagreement here. Nothing else in this manuscript depends on that comparison, since the three specifications are compared against one another within a single pipeline under common random numbers.

3.3 Power by analysis specification

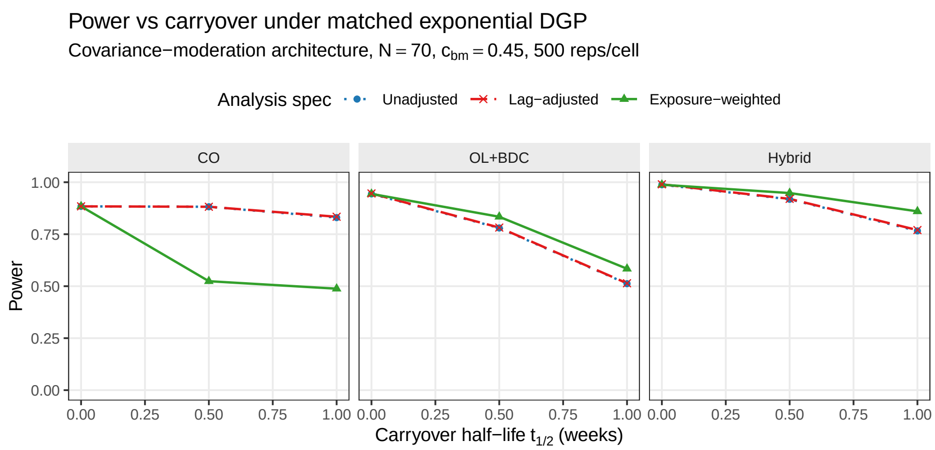

Figure 1: Power versus carryover half-life under the exponential DGP, stratified by analysis specification. Unadjusted and Lag-adjusted are indistinguishable throughout, their curves overlapping in all three panels. Exposure-weighted retains a modest advantage in the Hybrid and OL+BDC designs at $t_{1/2} = 1.0$ weeks, but is markedly inferior under the crossover (CO) design.

Two features of Figure 1 carry the substance of this study.

The first is that the Unadjusted and Lag-adjusted curves lie on top of one another in every panel and at every half-life. At the reference cell they give 0.766 and 0.770 at $t_{1/2} = 1.0$ weeks, a gap of 0.004 against an MCSE of 0.018. Across all 216 cells in scope the mean difference is 0.003, only three cells differ by more than 0.02, and mean bias (0.372 versus 0.373), mean squared error (1.876 versus 1.885) and coverage (0.938 versus 0.936) agree to three decimal places. Since the two share an estimand, are fitted to the same simulated datasets, and differ by exactly one nuisance term, this is as clean a null as the design can produce: the lagged-treatment term contributes nothing on any measure. Section 4.1 takes up why.

The second is that the three specifications give nearly identical power at $t_{1/2} = 0$, where there is no carryover to model, and diverge as the half-life increases, with Exposure-weighted separating from the other two. It retains a modest but consistent advantage in the Hybrid and OL+BDC designs, first appearing at $t_{1/2} = 0.5$ weeks and growing through $t_{1/2} = 1.0$ weeks. That advantage does not appear under the crossover design, where Exposure-weighted is in fact markedly inferior to both alternatives, 0.488 against 0.830 for Unadjusted at the matched cell. The mechanism appears straightforward: in a crossover trial the within-subject correlation already carries most of the signal that the decaying predictor was designed to recover, so it has little left to gain, and its down-weighting of the off-drug observations only works against it. We return to this in Section 4.

3.4 Decay-form mis-specification

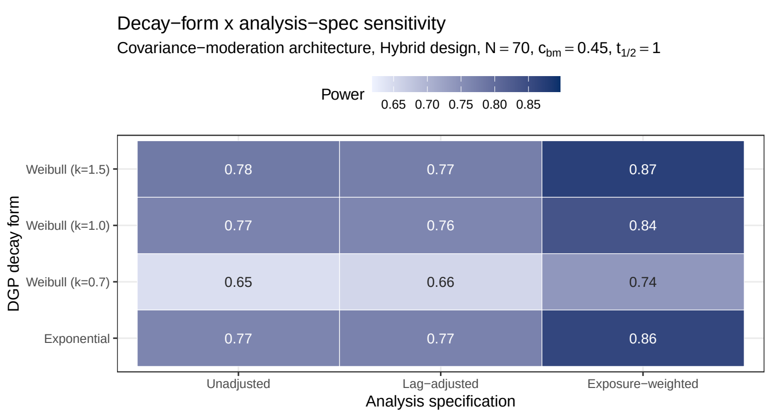

Figure 2: Power at the critical cell ($t_{1/2} = 1.0$ weeks, Hybrid design, $N = 70$, $c_{bm} = 0.45$) across DGP decay forms (rows) and analysis specifications (columns). Under the heavy-tailed Weibull DGP ($k = 0.7$), the Exposure-weighted advantage narrows because its assumed exponential decay no longer matches truth. The Unadjusted and Lag-adjusted columns are near-identical throughout. The fill scale is stretched to the observed power range, not the full [0,1] range, so that cell-to-cell differences are visible; the absolute shading should not be compared to other figures in this manuscript.

The heatmap cell at the intersection of the exponential DGP row and the Exposure-weighted column is the matched reference, with power 0.860 ± 0.016. The Unadjusted and Lag-adjusted columns are invariant to the assumed decay form by construction, since neither consumes one, so variation down those columns reflects only the data-generating decay and Monte Carlo noise; the two columns are near-identical to each other throughout, as in Figure 1. The remaining rows of the Exposure-weighted column give the cost of assuming exponential decay in the analysis when the true decay form is Weibull. Even under the heaviest-tailed data-generating process we examined ($k = 0.7$), the Exposure-weighted advantage narrows but does not reverse. We take up the practical interpretation of this in Section 4.

3.5 Type I error control and cluster-robust recalibration

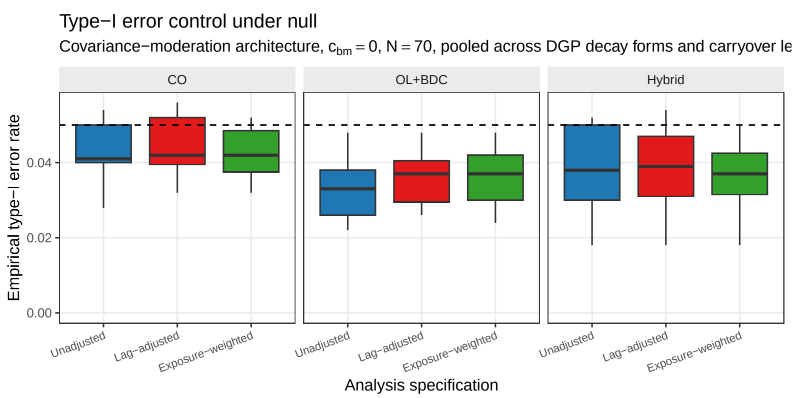

Figure 3: Empirical Type I error rates under the null ($c_{bm} = 0$), pooled across DGP decay forms and carryover half-lives, stratified by design and analysis specification. Boxplots cover the distribution across cells at $N = 70$; the dashed reference line marks the nominal 0.05 level.

Across the null cells, empirical Type I error sits uniformly below the nominal 0.05 level for all three specifications and in all three designs, with design-level means of 0.032 to 0.041. This shortfall is not sampling noise but the conservatism of the model-based interaction test described in Section 2.6, arising from a working covariance that does not match the one the data-generating mechanism induces. Because the conservatism is uniform across specifications, it does not disturb the comparison among them, and the cluster-robust recalibration reported below (block S6) confirms that a CR2 cluster-robust standard error restores the rejection rate to nominal without changing which specification leads. This also establishes that the null result of Section 3.3 is not an artifact of differential test size: Unadjusted and Lag-adjusted are conservative to the same degree,

Table 3: Conservatism of the model-based interaction test and its cluster-robust correction, at the null cell ($c_{bm} = 0$, exponential DGP, $t_{1/2} = 1.0$ weeks, Hybrid design, $N = 70$, the covariance-moderation architecture, 3,000 replicates). κ is the calibration ratio defined in Section 2.6; $\kappa > 1$ indicates a conservative test. All three specifications are conservative under the model-based test, most so Exposure-weighted ($\kappa = 1.10$, Type I = 0.029); the CR2 bias-reduced cluster-robust standard error restores κ to near one and Type I to nominal (0.045 to 0.049) for all three.

Specification	κ model	Type I model	κ CR2	Type I CR2
Unadjusted	1.06	0.036	0.98	0.048
Lag-adjusted	1.05	0.039	0.98	0.049
Exposure-weighted	1.10	0.029	1.00	0.045

Table 4: Cluster-robust recalibration at the reference cell and four stress cells (500 replicates, $c_{bm} = 0.45$, the covariance-moderation architecture). EW is Exposure-weighted, LA is Lag-adjusted, U is Unadjusted. κ is the calibration ratio of Section 2.6, here for Exposure-weighted; a value above one is a conservative test. The two gap pairs give each alternative against the unadjusted benchmark under each inference. CR2 df is the mean Satterthwaite degrees of freedom for the cluster-robust test on Exposure-weighted. Computed with `clubSandwich`.

Cell	κ model	κ CR2	EW model	EW CR2	EW−U model	EW−U CR2	LA−U model	LA−U CR2	CR2 df
Reference, $N = 70$	1.15	0.99	0.829	0.863	+0.080	+0.092	-0.003	+0.002	23
Small $N = 35$	1.13	0.96	0.558	0.590	+0.066	+0.086	-0.002	+0.000	12
High autocorr, $\rho = 0.9$	1.26	0.94	0.954	0.980	+0.052	+0.032	+0.000	-0.004	23
Half-life mis-spec	1.14	1.00	0.759	0.795	+0.010	+0.024	-0.003	+0.002	23
30% MCAR dropout	0.99	0.85	0.148	0.126	+0.010	+0.010	-0.004	+0.010	5

so their equal power reflects equal information rather than an offsetting calibration difference. Table 3 makes the per-specification conservatism, and its cluster-robust correction, explicit at the null cell.

Figure 4 shows the effect of recalibration on test size directly. The model-based null rejection rate sits below nominal across all five S6 cells and all three specifications, while the CR2 test restores it to within the Monte Carlo band around 0.05. This matters because a test that is not correctly sized cannot fairly be preferred on power alone. In the information-rich cells, the calibration ratio κ runs from 1.07 to 1.26 under the model-based test and is restored to about one under CR2, and the correctly sized test recovers two to five points of power in the process.

Table 4 reports both alternatives against the unadjusted benchmark. The Exposure-weighted advantage is held or slightly widened under CR2 at the reference cell (+0.080 to +0.092), at the small-N cell (+0.066 to

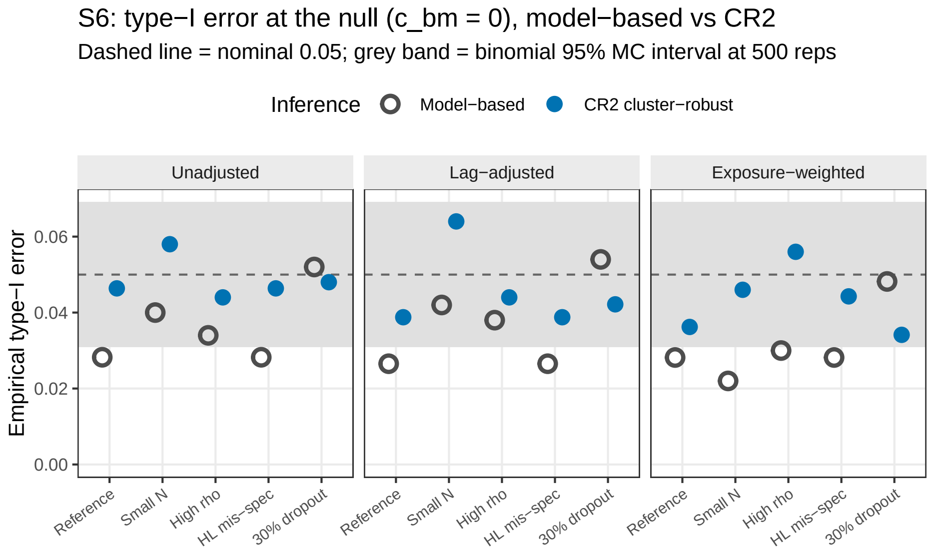

Figure 4: Empirical Type I error at the null ($c_{bm} = 0$) under model-based and CR2 cluster-robust inference, across the five S6 stress cells and the three specifications. The model-based test (open) is conservative (near 0.03); CR2 (filled) is restored to the nominal 0.05 (dashed), within the binomial Monte Carlo band (gray). 500 null replicates per cell.

+0.086), and at the half-life mis-specification cell (+0.010 to +0.024); it narrows under high autocorrelation (+0.052 to +0.032, though it still leads); and at 30% MCAR dropout it is small under either inference (+0.010). We read the dropout cell as a restatement, from a different angle, of the S3 finding below: the advantage is driven by information carried in the off-drug observations, and once dropout has removed most of that information there is little left to distinguish. The Lag-adjusted gap, by contrast, is within ±0.004 of zero at every cell under the model-based test and within ±0.010 under CR2, which corroborates the null of Section 3.3 under a second inference procedure and at four stress cells the main grid does not cover. This recalibration was validated only in the Hybrid, $N = 70$ cells, and it should not be extended to the crossover design, where Exposure-weighted does not lead in the first place.

3.6 Sensitivity analyses

Four Tier 2 blocks perturb one axis at a time around a common reference cell (Hybrid, $N = 70$, $c_{bm} = 0.45$, exponential DGP, $t_{1/2} = 1.0$), holding the remaining factors fixed. The blocks are reported separately here and are not pooled with Tier 1 or with each other.

3.6.1 S1: within-factor autocorrelation

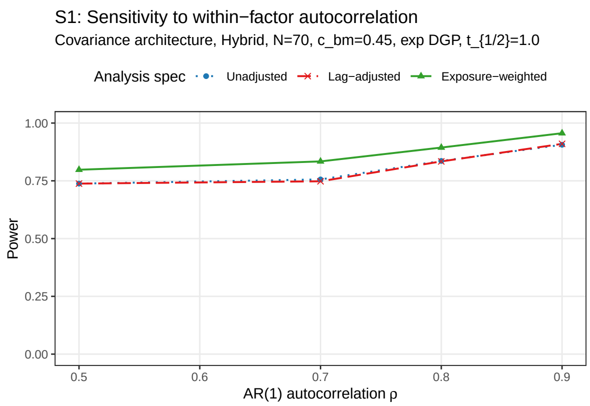

Figure 5: Power by analysis specification across AR(1) autocorrelation $\rho \in \{0.5, 0.7, 0.8, 0.9\}$.

All three specifications gain power monotonically with ρ. At $\rho = 0.5$, Exposure-weighted gives 0.798 ± 0.018 against 0.738 ± 0.020 for both Unadjusted and Lag-adjusted, which are identical to three decimal places at this level; at $\rho = 0.9$, Exposure-weighted reaches 0.956 ± 0.009 against 0.906 ± 0.013 and 0.910 ± 0.013 (point estimate plus or minus MCSE, $n_{sim} = 500$). The Exposure-weighted advantage, roughly 0.05 to 0.08 in absolute power throughout, is preserved at every autocorrelation level we examined, which suggests that autocorrelation strength and the choice of specification act additively rather than interacting. Lag-adjusted tracks Unadjusted across the whole range, never differing by more than 0.008.

3.6.2 S2: analyst-truth half-life mismatch

Block S2 crosses two true half-lives ($t_{1/2} \in \{0.5, 1.0\}$ weeks) with four analyst-assumed values ($\hat{t}_{1/2} \in \{0.25, 0.5, 1.0, 2.0\}$ weeks), giving eight cells. Only Exposure-weighted consumes $\hat{t}_{1/2}$; the other two are invariant to it by construction. The cost of a wrong assumption must therefore be read within a true half-life rather than pooled across the block, since most of the variation across all eight cells is driven by the true half-life itself and not by the analyst's error.

Read that way, the penalty is modest. At a true half-life of 1.0 weeks, Exposure-weighted attains 0.832 when correctly specified, and 0.766, 0.808 and 0.826 when the assumed value is 0.25, 0.5 and 2.0 weeks respectively; the worst case, a fourfold under-assumption, costs 6.6

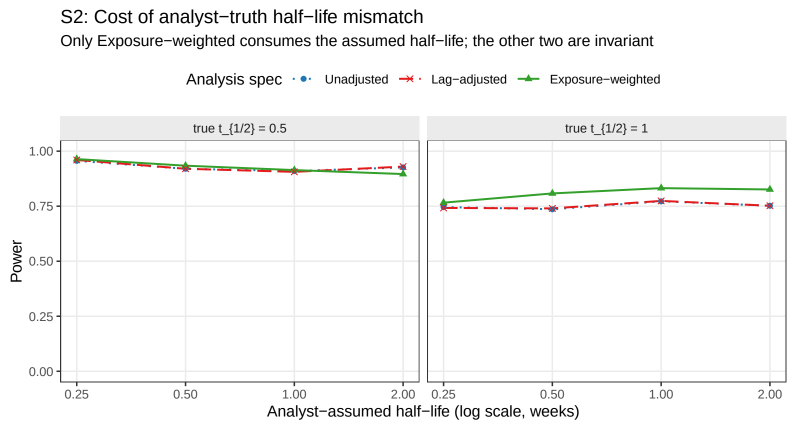

Figure 6: Power as the analyst's assumed half-life departs from the true DGP half-life. Only Exposure-weighted consumes the assumed half-life; Unadjusted and Lag-adjusted are invariant to it by construction and appear as flat reference lines.

percentage points and still leaves it above both alternatives (0.746 and 0.742 at that cell). At a true half-life of 0.5 weeks it attains 0.934 when correctly specified, against 0.964, 0.914 and 0.896 at the same four assumed values.

Two features of that second row deserve comment, and the block supplies its own calibration for both. Because Unadjusted and Lag-adjusted cannot depend on $\hat{t}_{1/2}$, their variation along that axis is pure Monte Carlo error, and it is not small: at a true half-life of 0.5 weeks they range over 0.910 to 0.956 and 0.906 to 0.960 respectively across the four columns. The block's noise floor is therefore roughly four to five percentage points, and differences below it should not be interpreted. This disposes of the apparent anomaly that assuming 0.25 weeks (0.964) beats correct specification (0.934), a 3.0-point difference well inside that floor. It also qualifies the second concern: at a true half-life of 0.5 weeks with an assumed value of 2.0 weeks, Exposure-weighted falls to 0.896, below both alternatives at 0.926 and 0.930, so a fourfold over-assumption erodes its advantage to a tie rather than preserving it. The direction is worth noting even though the magnitude is not resolvable here.

The practical reading is that an approximate half-life is adequate, with an asymmetry: under-assuming is safer than over-assuming. That follows from the limiting-case relationship of Section 2.4, since shrinking $\hat{t}_{1/2}$ drives the predictor continuously toward the binary indicator and hence toward Unadjusted, which remains a serviceable specification, whereas inflating it drives the predictor toward a constant, at which the on-off contrast and with it the estimand itself degenerate.

3.6.3 S3: dropout

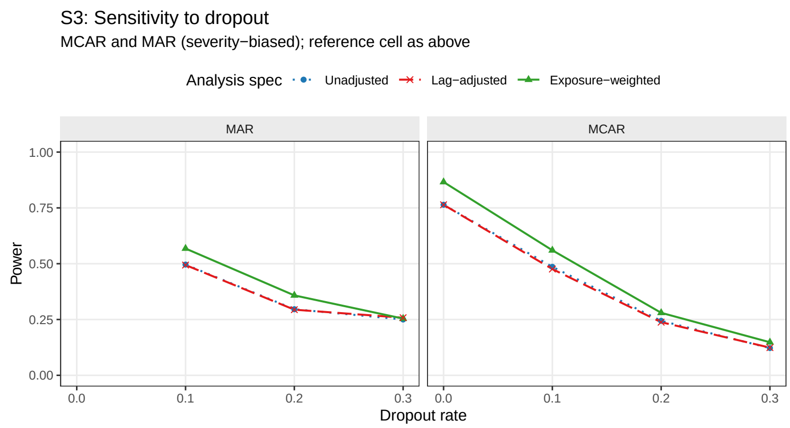

Figure 7: Power at dropout rates $\in \{0, 0.1, 0.2, 0.3\}$ under MCAR and MAR mechanisms.

All three specifications lose power monotonically as the dropout rate increases. At the MCAR baseline ($N = 70$, no dropout), Exposure-weighted gives 0.866 ± 0.015 while Unadjusted and Lag-adjusted both give 0.764 ± 0.019; at an MCAR dropout rate of 0.30, these collapse to 0.148 ± 0.016, 0.122 ± 0.015 and 0.124 ± 0.015 respectively. The Exposure-weighted advantage narrows steadily as dropout worsens: the absolute gap over Unadjusted runs about 0.10 at zero dropout, 0.07 at a rate of 0.10, 0.04 at 0.20, and 0.03 at 0.30 under MCAR, narrowing further under MAR dropout at the highest rate examined. This pattern is consistent with that advantage originating in the exposure-weighted contrast contributed by off-drug observations; when those observations are preferentially lost, the edge contracts. At low to moderate dropout, that is, at a rate of 0.20 or below, the advantage is preserved. Lag-adjusted again tracks Unadjusted at every dropout rate and under both mechanisms, the largest discrepancy being 0.010.

3.6.4 S4: biomarker-moderation effect-size curve

The power curve is sigmoidal for all three specifications, as expected. At $c_{bm} = 0$, empirical Type I error sits below nominal for each, at $0.026 \pm$

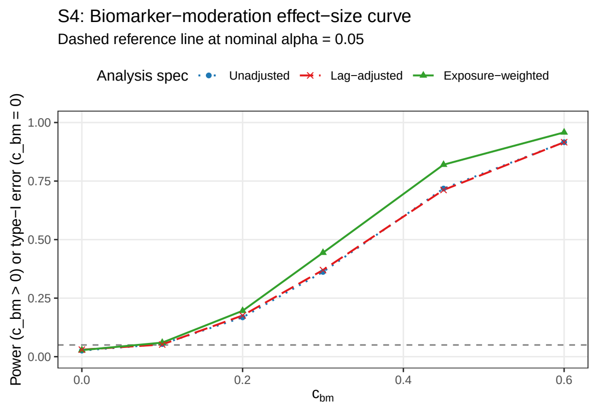

Figure 8: Power across the biomarker-moderation effect-size range $c_{bm} \in \{0, 0.10, 0.20, 0.30, 0.45, 0.60\}$. The $c_{bm} = 0$ point is the null for Type I verification.

0.007, 0.030 ± 0.008 and 0.028 ± 0.007 for Unadjusted, Lag-adjusted and Exposure-weighted respectively; power then rises through the mid-range and saturates once $c_{bm} \geq 0.45$, reaching 0.916 ± 0.012, 0.916 ± 0.012 and 0.958 ± 0.009 at $c_{bm} = 0.6$. The Exposure-weighted advantage over the unadjusted benchmark is largest at intermediate effect sizes, $+0.082$ absolute at $c_{bm} = 0.30$ and $+0.102$ at $c_{bm} = 0.45$, shrinking to $+0.042$ at $c_{bm} = 0.60$ as all three approach the ceiling. The practical implication is that the advantage matters most in trials targeting a moderate biomarker-treatment interaction, and matters least when the true effect is either very small or comfortably large. Lag-adjusted and Unadjusted coincide across the entire effect-size range, differing by at most 0.008 at any point on the curve.

3.7 A higher-precision check

Because a number of the Tier 1 gaps are small, we reran a focused 24-cell grid (OL+BDC design, $N \in \{35, 70\}$, true half-life $\in \{0.5, 1.0\}$) at 600 trials per cell, all converging, and compared specifications with a paired McNemar test, considerably more sensitive than an independent-groups comparison since all three are fit to the same trials. Type I error across the twelve null cells fell in $[0.003, 0.053]$, so nominal alpha holds. Paired McNemar tests place Exposure-weighted ahead of the unadjusted benchmark at all four alternative cells, by $+2.3$ to $+13.7$ percentage points, with the largest gap at the highest-leverage cell ($N = 70$, true $t_{1/2} = 1.0$ weeks, $p = 6.6 \times 10^{-17}$) and $p \leq 0.015$ at the remaining three, a sharper confirmation than the Tier 1 grid alone provides.

One scope caveat governs the rerun's third arm. It was produced by the package's `lme_analysis()` routine, whose carryover option appends time since discontinuation as a linear main effect rather than the lagged indicator L_{it} used in the Tier 1 grid. It is therefore not the Lag-adjusted specification of Section 2.4 and is not reported as such here; that alternative remains a reasonable specification we have not evaluated (Section 4.5).

4 Discussion

4.1 The lagged-treatment term does no work

The clearest result of this study is a negative one. Adding the classical lagged-treatment indicator to the analysis model changed nothing measurable, anywhere in the grid. Across all 216 cells the mean power difference against the unadjusted benchmark was 0.003 against a Monte Carlo standard error of 0.018; bias, mean squared error and coverage agreed to three decimal places; and the equality held in every sensitivity block, at every autocorrelation level, every dropout rate and every effect size, and at all five cluster-robust stress cells under both inference procedures.

The strength of this conclusion rests on the design rather than on the size of the grid. The two specifications estimate the same coefficient, were fitted to the same simulated datasets under common random numbers, and differ by exactly one term. There is no confounding from a change of estimand, from Monte Carlo variation between arms, or from differential test size, since Section 3.5 shows both to be conservative to the same degree. What is left is the term itself, and it contributed nothing.

The mechanism follows from where the term sits. Carryover damages the interaction test on two fronts, as set out in Section 1: it attenuates the coefficient, because contaminated occasions are coded as drug-free, and it inflates the residual variance, because those occasions are heterogeneous in a way the model does not represent. The lagged indicator enters only as a nuisance main effect, with no bm:L_{it} product term, so the interaction it tests remains bm:D_{it} and the attenuation is untouched. It can in principle absorb some of the second effect, and in these designs it evidently absorbs too little of it to matter: the indicator marks a single occasion per randomization path and is constant thereafter, so it cannot describe a graded decline across a washout that runs two to four occasions.

A natural objection is that this null reflects the crudeness of the particular device rather than the futility of the approach. The lagged indicator marks one occasion and is constant thereafter, so it cannot describe a graded decline, and a more capable nuisance term might succeed where it fails. We therefore ran a targeted check on the alternative that the reference implementation itself provides, which appends time since discontinuation as a linear main effect, `Sx ~ bm + t + Db + bm:Db + tsd`, and which we call the washout-adjusted specification (stored as `E4`). Unlike the lagged indicator it is indexed by elapsed time rather than by position in the visit sequence, so it can in principle track the whole washout. The check covers the three designs crossed with true half-lives of 0.5 and 1.0 weeks at $N = 70$, $c_{bm} = 0.45$ under the exponential DGP, at 500 replicates, with all four specifications fitted to the same datasets.

It does slightly better, and the improvement is instructive rather than material. Empirical Type I error is at nominal, spanning 0.030 to 0.070 with a mean of 0.050 against 0.030 to 0.064 and a mean of 0.045 for the unadjusted benchmark, so the comparison is not contaminated by differential test size. Power exceeds the benchmark in all six cells, by a mean of 1.3 percentage points (sign test $p = 0.031$), and paired McNemar tests reach significance under the crossover design at both half-lives (+1.8 points, $p = 0.008$; +2.2 points, $p = 0.003$) and under OL+BDC at the longer half-life (+2.2 points, $p = 0.015$), but not under the Hybrid design (+1.0 points, $p = 0.27$).

The mechanism is exactly the one Section 2.4 predicts, and it explains why the gain is confined where it is. Point estimates are unchanged from the unadjusted values, so nothing about the attenuation is repaired; the entire effect runs through the standard error, whose ratio to the unadjusted one is 0.959 to 0.964 under the crossover design, 0.978 to 0.989 under OL+BDC, and 1.000 under Hybrid. A term indexed by elapsed time has something to describe only when the washout is long enough to vary, and the designs differ sharply in this respect: the crossover schedule places four off-drug occasions between 2.5 and 10 weeks after discontinuation, whereas the Hybrid washouts run one to two weeks. The practical ceiling on what a carryover nuisance term can deliver is therefore about two percentage points, and only in designs with a long post-discontinuation window, against the nine to eleven points that exposure weighting delivers where it works.

Two limits bound this check. It covers only the exponential DGP at a single sample size and effect size, so it is a probe rather than a fourth arm; and it shares the structural limitation of every specification evaluated here, entering carryover as a main effect only (Section 4.5).

For practitioners the implication is direct. A carryover nuisance term of either form, lagged or washout-adjusted, should not be expected to recover meaningful power lost to carryover, and reporting one does not constitute an effective adjustment for it.

4.2 The exposure-weighted advantage, and where it fails

The exposure-weighted predictor does help, but conditionally. At its best cell (Hybrid design, $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1.0$) it exceeds the unadjusted benchmark by about 10 percentage points of power (0.860 versus 0.766), a difference of roughly five Monte Carlo standard errors and therefore not attributable to chance, and the high-precision rerun of Section 3.7 confirms an advantage at all four of its cells. On the other hand, under the crossover design it performs considerably worse (0.488 versus 0.830). The reason for this reversal appears straightforward: in a crossover trial, the within-subject correlation already carries most of the signal that the decaying predictor was designed to recover, so it gains little from its more elaborate representation of exposure, and its down-weighting of off-drug observations works against it. The recommendation therefore holds only in the two non-crossover designs studied here, and should not be extended to a crossover trial, where the unadjusted binary indicator is materially better.

That the one specification which addresses attenuation is also the only one to gain power, while the one which addresses only residual variance gains nothing, suggests that attenuation is the mechanism that matters in this setting. We offer that as an interpretation consistent with the results rather than as a finding, since the design does not isolate the two channels independently.

4.3 What the robustness checks add

Three checks probe whether the exposure-weighted advantage, where it exists, is fragile. On decay shape (Figure 2): even when the analyst incorrectly assumes exponential decay and the true form is Weibull, it still does not fall below the unadjusted benchmark, within the range of decay shapes we examined. On the assumed half-life (Figure 6): the other two specifications consume no half-life and are unaffected, while this one is, yet at a true half-life of one week mis-specifying it by a factor of four moves power from 0.832 down to 0.766 at worst, a penalty of 6.6 percentage points that still leaves it at or above the alternatives. The exception is over-assumption at a short true half-life, where its advantage erodes to a tie (Section 3.6). On autocorrelation (Figure 5): stronger serial correlation raises all three specifications by roughly the same amount, so the lead of five to eight points holds regardless of correlation strength. Taken together, these checks suggest that where the advantage exists, it is not a fragile artifact of one particular combination of nuisance parameters.

One caveat qualifies the Tier 1 figures throughout. Outside block S2 the analysis-side half-life is set equal to the data-generating one, so the exposure-weighted specification is granted an oracle no real analyst possesses. Its Tier 1 margins are therefore upper bounds, and block S2 gives the more realistic figure.

4.4 Dropout matters more than the method

The finding with the largest practical consequence in this study concerns missing data, not carryover. Unfortunately, when patients drop out (Figure 7), power collapses for both specifications together: at 30% dropout, both fall to somewhere between 0.12 and 0.25 and become essentially indistinguishable. No specification offers protection against this loss; each can only organize whatever data happen to remain. Dropout tied to symptom severity (MAR) is somewhat worse than dropout occurring at random (MCAR), because the sickest patients, who carry the most signal, tend to leave first. The practical implication is direct: retaining patients in the trial yields considerably more statistical power than any choice among these three specifications ever could.

4.5 Scope, related work, and limits

This manuscript deliberately narrows the full-scope grid to the covariance-moderation architecture and to the non-linear decay forms, in the interest of a self-contained evaluation of manageable length. The archived full-scope drafts (`archive/report.Rmd` and `archive/report-slim.Rmd`) show that the exposure-weighted advantage documented here does not carry over to mean moderation, where the biomarker moderates the mean response directly and Exposure-weighted is marginally dominated in every design examined; readers who need that cross-architecture comparison should consult those drafts rather than extrapolate from this one. Establishing what the originally published power values (Section 3.2) were computed under remains outstanding work.

The crossover tradition surveyed by Jones and Kenward[4] would lead one to expect a shape-free lagged term to outperform a parametric predictor once the decay shape is mis-specified. Our results do not support that expectation over the range examined, for two reasons: the exposure-weighted penalty for an incorrect half-life is too small to be overtaken (Section 3.6), and more fundamentally the lagged term does not improve on ignoring carryover at all (Section 4.1), leaving no advantage for decay-shape mis-specification to erode.

One adjacent specification was not evaluated, and it bounds the negative result of Section 4.1. Every carryover term considered here, lagged and washout-adjusted alike, enters as a nuisance main effect with no product against the biomarker, so our finding is that a carryover *main effect* does not recover the attenuated signal, not that no shape-free specification can. The natural candidate is a carryover term interacted with the biomarker, either `Sx ~ bm + t + Db + L + bm:Db + bm:L` or its washout-adjusted counterpart with `bm:tsd`, letting the biomarker effect decline across the washout without committing to a rate; the second, a linear approximation to what exposure weighting does parametrically, is the more promising candidate and the obvious next step.

Two further limitations bound what we can conclude. The numbers presented here come from a single parameter set calibrated to a prazosin and PTSD trajectory, so absolute power values should be read as illustrative and the ranking, itself conditional on trial design, as the more portable result. And the sensitivity checks vary one factor at a time and can therefore miss genuine interactions between factors, for instance between high autocorrelation and heavy dropout; the one joint check available to us, block S5 in the full-scope manuscript, revealed no such interaction.

5 Conclusions

1. **The lagged-treatment term does nothing.** Adding the classical lagged just-off-drug indicator never improved on ignoring carryover altogether, on any measure, in any design, at any half-life, and in every sensitivity block. Because the two specifications share an estimand, were fitted to the same datasets, and differ by a single term, this is a clean null rather than an underpowered comparison. We trace it to the term entering as a nuisance main effect, which leaves the attenuation of the interaction untouched. A lagged term of this form should not be reported as an adjustment for carryover.

2. **Exposure weighting helps, but conditionally.** The exposure-weighted predictor achieves the highest power, about 10 points over the unadjusted benchmark, only in the Hybrid and OL+BDC designs, and its lead survives an incorrectly assumed decay shape and an incorrectly assumed half-life within those designs. Under the crossover design it is markedly worse than the binary indicator (0.488 against 0.830) and should not be used. Where it is used, we recommend reporting it with a CR2 cluster-robust standard error (Block S6).

3. **An approximate half-life is adequate where exposure weighting is used, but err short.** At a true half-life of one week, an assumed value wrong by a factor of four costs 6.6 percentage points of power, so a rough pharmacokinetic estimate is sufficient in practice. The error is asymmetric: under-assuming drives the predictor toward the binary indicator, which remains serviceable, whereas over-assuming drives it toward a constant and erodes the advantage to a tie. Note also that outside block S2 the analyst is granted the true half-life, so the Tier 1 margins are upper bounds.

4. **Dropout matters more than the choice of method.** Power falls sharply as dropout increases, roughly halving at a dropout

rate of 20%, and this loss affects all three specifications equally. Preventing dropout is a more effective use of design effort than any choice among them.

5. **The interaction test is somewhat conservative, and the conservatism can be corrected.** All three specifications reject slightly less often than the nominal 5% under the null, and a CR2 cluster-robust standard error restores the correct rate while recovering a modest amount of power. The exposure-weighted advantage over the benchmark is held or widened under CR2 at four of the five S6 stress cells, and is small under 30% dropout, consistent with the fourth conclusion above (Block S6).

6. **This is a narrowed slice of a larger study.** The mean-moderation architecture and linear decay fall outside the scope of this manuscript; the full-scope grid reported in the archived drafts `archive/report.Rmd` and `archive/report-slim.Rmd` is the authority on those comparisons, and should be consulted before generalizing beyond the covariance-moderation architecture. We also make no claim of numerical agreement with the originally published power values (Section 3.2).

6 Conflict of interest

The authors declare no conflicts of interest.

7 Funding

This work received no specific funding.

8 Data availability statement

The data and code supporting the findings of this study are available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/carryover-sensitivity/`. Exact paths for this manuscript are listed in the Reproducibility section below.

9 Reproducibility

This manuscript was rendered with R 4.6.0 inside the project's zzcollab Docker container. The production simulation ran under R 4.5.3 (§3.6), with the image hash in `Dockerfile` and the package set in `renv.lock`. The pipeline uses `nlme` for the continuous-time linear-mixed analysis, `parallel` for the 8-core parameter-grid sweep, and `data.table` or `furrr`/`purrr` in the two simulation collections; no simulation code runs during knitting, which loads pre-computed checkpoints only. Each driver sets a master seed (20260507) and derives per-replicate seeds from it via `parallel::mc.set.seed`.

Data and code availability. All simulation outputs, cell-level summaries, and driver scripts are versioned under `analysis/scripts/carryover-sensitivity/` and `analysis/data/` in the pmsimstats-ng repository, including the principal factorial, the S1 to S5 sensitivity blocks, the S6 cluster-robust recalibration, the washout-adjusted (E4) check of Section 4.1, and the high-precision prototype rerun of Section 3.7. The grid design and Morris ADEMP pre-registration are in `simulation-grid-plan.md` in the same directory. Superseded wider-scope drafts are retained under `archive/`, documented in its `README.md`.

Specification labels. The three specifications are stored as `E1`, `E2`, `E3` and displayed here as Unadjusted, Lag-adjusted, and Exposure-weighted. The mapping is applied only at display time, by `spec-labels.R`, so the stored values and archived outputs are unaffected. Output archived before 2026-07-31 stores the specifications as `A1`, `A2`, `A3`; the migration is documented in `analysis/report/NOTATION.md`.

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